The conjectures of the revised proposal (chemistry prior, commitment, several dates): proofs, counterexamples and corrected statements

Second companion note

31 October 2026

Abstract

The revised proposal keeps the benchmark model (M2) and adds two relaxations: a two-class chemistry prior (M3) and commitment (M4). It states five conjectures. Conjectures 4 and 5, and the benchmark part of Conjecture 1, are unchanged; the first companion note settles them, and its results stand. This note settles the rest.

- *Conjecture 1 on the chemistry prior is false*, for two separate reasons. The after-date decision can make information pivotal, as in the benchmark. More strikingly, *any* sting breaks threshold rules once a relationship is valuable enough. A refusal is most informative at intermediate beliefs, so the agent asks when fairly sure the chemistry is low, and when fairly sure it is high, but not in between. Without a sting, and with an after-date plan that does not depend on beliefs, threshold rules hold at every step.
- *Conjecture 2 is true at certainty and false in general.* With a known high-chemistry match, a higher relationship value always raises the value of escalating. At any fixed belief short of certainty it can lower it, when commitment is weak relative to the fear of bad news; an exact asymptotic slope says which. The claim that the range of beliefs with negative commitment value shrinks with R holds whenever the after-date plan does not switch. It fails, though only by small amounts, when it does.
- *Conjecture 3 is false* in the several-dates model built on the chemistry prior, solved exactly by value iteration. A better outside option lowers the threshold for continuing at beliefs where the agent would ask for another date. Its Phase C part is true.

Two further findings bear on the proposal. Under the chemistry prior with a sting, even a monotone signal about chemistry can have negative value. And Proposition 6 needs an extra condition. Every number is reproduced by `verify_v2.py` (model code in `chem.py`).

Summary

Claim	Verdict	What is proved
Conjecture 1, chemistry prior	False (Props. 4, 5)	Thresholds in π at every step without a sting, when the after-date plan is the same at every belief (Theorem 3). With a sting, the last-step rule is "ask iff $A \geq \rho(\pi)$", with ρ zero at $\pi = 0, 1$ and positive in between. For large R this yields two asking bands, near 0 and near 1 (Proposition 5).
Conjecture 1, benchmark	Unchanged	First note: false in general, true under condition (M).
Conjecture 2, first part	True at $\pi = 1$; false at fixed $\pi < 1$ in general	At $\pi = 1$ the gain from a yes is nondecreasing in R, under any stings (Theorem 7). For large R, $\partial D/\partial R = \Phi(\pi)$ exactly, with $\Phi > 0$ near $\pi = 1$ and possibly $\Phi < 0$ elsewhere (Proposition 8).
Conjecture 2, second part	True without plan switches; false in general	The negative-commitment set shrinks with R at beliefs where the after-date plan does not switch, and always at $\pi = 0, 1$. A counterexample uses a switch; violations found are small (Proposition 10).
Conjecture 3 (M2+)	Phase C true; on the day false	Exact full-recursion counterexample, with no sting (Proposition 12).
Conjectures 4, 5	Unchanged	First note's results stand. The threshold part of Conjecture 4 extends to M3/M4 when refusals are bad news. The spend part of Conjecture 5 loses its guarantee under the chemistry prior with a sting (Section 5).

1 Setting

Notation follows the revised proposal. There are two classes, $z \in \{L, H\}$, with $(C, q, q_D) = (C_z, q_z, q_{D,z})$, $C_H > C_L$, $q_H > q_L$ and $q_{D,H} > q_{D,L}$ (A13). The belief is $\pi = P(z = H)$, and π_z denotes the weight on class z. At step j write

$$\bar{q} = \pi q_H + (1 - \pi) q_L, \qquad p = \varphi_j \bar{q}, \qquad \pi_{\text{yes}} = \frac{\pi q_H}{\bar{q}}, \qquad \pi_{\text{no}} = \frac{\pi(1 - \varphi_j q_H)}{1 - p}.$$

A short calculation gives

$$\pi_{\text{yes}} - \pi_{\text{no}} = \frac{\pi(1 - \pi)(q_H - q_L)}{\bar{q}\,(1 - \varphi_j \bar{q})}. \tag{1}$$

The effective weight is $\hat{p}(p) = p - \kappa_m(p)$, as in the first note.

Standing assumption S1. Ratings after step j are $\text{Bin}(m_j, C_z)$ given the class, with exact Bayes updating; this is MLRP in the class. The value after the date is truncated after one more date (21), except in Section 4. The relationship chance is $h_z(n)$ after a date that reached step n: $h_z(n) = h_z$ in M3, $\psi_n h_z$ (bonding) or $h_z + \theta_n$ (sliding) in M4, with $h_z = \bar{h}\, C_z q_{D,z}$. Another date has a fixed chance h_z^D. The expected sting is $\kappa_m(p)\Delta^+$.

Standing assumption S2. Where stated, the ask for another date carries no sting. This is the condition of the proposal's Propositions 4–8.

For a Phase-C plan $d \in \{\text{cut}, \text{ask}\}$, let X_z^d be its value against class z:

$$X_z^{\text{cut}} = U, \qquad X_z^{\text{ask}} = q_{D,z} V_{D,z} + (1 - q_{D,z})U, \qquad V_{D,z} = \delta_D\big(h_z^D C_z R + (1 - h_z^D)U\big).$$

Lemma 1 (After-date value)**.** *Under S1 and S2,*

$$V_{\text{end}}(\pi; n) = \max_d \sum_z \pi_z \Big[h_z(n) C_z R + \big(1 - h_z(n)\big) X_z^d \Big].$$

This is convex and piecewise linear in π. It is affine wherever the optimal plan is the same at every belief.

Proof. Bayes' rule after "no relationship" gives $(1-\mathbb{E}h)\,f(\pi_{\text{not}}) = \sum_z \pi_z(1-h_z(n))f(\delta_z)$ for every f that is linear in the belief. The value of each plan is linear in the belief once the sting is absent, so $(1-\mathbb{E}h)V_C(\pi_{\text{not}}) = \max_d \sum_z \pi_z(1-h_z(n))X_z^d$. □

Lemma 2 (Anatomy of an ask)**.** *At step j, with n steps completed, let X be the value after a yes (step, rating, continuation), and let*

$$G = X - V_{\text{end}}(\pi; n), \qquad L = V_{\text{end}}(\pi; n) - V_{\text{end}}(\pi_{\text{no}}; n), \qquad \Delta = G + L.$$

The value of asking minus ending is $D = pG - (1-p)L - \kappa(p)\Delta^+$. Suppose $V_{\text{end}}(\cdot; n)$ is affine with slope a_1, and write

$$A = X - V_{\text{end}}(\pi_{\text{yes}}; n), \qquad \Gamma = a_1(\pi_{\text{yes}} - \pi_{\text{no}}).$$

Then

$$D = p\,A - \kappa(p)\,(A+\Gamma)^+. \tag{2}$$

Proof. Bayes plausibility gives $p\pi_{\text{yes}} + (1-p)\pi_{\text{no}} = \pi$, so an affine V_{end} satisfies $pV_{\text{end}}(\pi_{\text{yes}}) + (1-p)V_{\text{end}}(\pi_{\text{no}}) = V_{\text{end}}(\pi)$. Hence $pG - (1-p)L = pA$. Also $\Delta = X - V_{\text{end}}(\pi_{\text{no}}) = A + \Gamma$. □

Here A is the value of the step net of the news a yes carries, and Γ is the gap the news opens between a yes and a no. By (1), Γ is zero at $\pi = 0$ and $\pi = 1$ and positive in between.

2 Conjecture 1 on the chemistry prior

Theorem 3 (Thresholds without a sting)**.** *Under S1 and S2 in M3, suppose there is no sting on the day ($\eta = 0$) and the optimal Phase-C plan is the same at every π. Then for every step, $D_j = p_j A_j$ with A_j nondecreasing in π. So the agent asks exactly on $\{\pi \geq \pi_j^*\}$.*

Proof. By Lemma 1, V_{end} is affine, and Lemma 2 with $\kappa = 0$ gives $D_j = p_j A_j$. Since the rating is a martingale on the affine V_{end},

$$A_j(\pi) = W_j^*(\pi_{\text{yes}}) - c_j + \mathbb{E}\big[O_{j+1}(\pi') \mid \pi_{\text{yes}}\big], \qquad O_{j+1} = D_{j+1}^+.$$

By induction, $O_{j+1} = (p_{j+1}A_{j+1})^+$ is nondecreasing: p is increasing and A_{j+1} is nondecreasing. The posterior after a rating is increasing in the prior and in the rating, and the rating's predictive distribution rises with the belief, so the expectation is nondecreasing in π_{yes}. W_j^* is increasing in $\mathbb{E}[C]$. Finally, π_{yes} is increasing in π. Since $p_j > 0$, D_j has the sign of A_j. □

Proposition 4 (Counterexample A: the after-date decision is pivotal)**.** *Take M3 with $C = (0.2, 0.8)$, $q = (0.895, 0.9)$, $q_D = (0.795, 0.8)$, $\bar{h} = 0.5$, $R = 6000$, $U = 1000$ and $\delta_D = 0.9$. The last step has $v = 50$ and $k = 30$ with one Bernoulli rating, and there is no sting anywhere. The agent asks on $[0.0681, 0.2340] \cup [0.6654, 1]$. The mechanism is that of the first note's Proposition 8, carried into M3.*

Proposition 5 (Counterexample B: fear of finding out)**.** *Under S1 and S2 in M3, take the last step ($O \equiv 0$), and suppose the Phase-C plan is constant, so V_{end} is affine with slope $a_1 > 0$. Then $A(\pi) = W^*(\pi_{\text{yes}}) - c$ is nondecreasing, and the agent asks if and only if*

$$\hat{p}(p) > 0 \quad \textit{and} \quad A(\pi) \;\geq\; \rho(\pi) := \frac{\kappa(p)}{\hat{p}(p)}\, a_1\, \frac{\pi(1-\pi)(q_H - q_L)}{\bar{q}(1-\varphi\bar{q})}. \tag{3}$$

Suppose further:

- $\kappa > 0$ *on* $(0,1)$*: D1 or D2 with* $\eta > 0$*, or KR with* $\eta(\Lambda - 1) > 0$*;*
- $\hat{p}(\varphi q_L) > 0$*;*
- *the step is worth taking even with a low-chemistry match:* $A(0) = W^*(C_L) - c > 0$.

Then there is $\bar{a}$ *such that, whenever* $a_1 > \bar{a}$*, the agent asks for all* π *near* 0 *and all* π *near* 1*, but not at* $\pi = 1/2$*. In particular, if* $\delta_D = 0$*, or if* $\delta_D h_L^D C_L > 0$*, then* $a_1 \to \infty$ *as* $R \to \infty$*, so the asking set fails to be a threshold for every* R *large enough.*

Proof. The rule (3). In (2), if $A + \Gamma > 0$ then $D = \hat{p}A - \kappa\Gamma$, which is non-negative iff $A \geq \kappa\Gamma/\hat{p}$ when $\hat{p} > 0$. If $A + \Gamma \leq 0$ then $D = pA \leq 0$. If $\hat{p} \leq 0$ and $A + \Gamma > 0$ then $D < 0$, because $\Gamma > 0$. Substitute (1).

Near 0 *and* 1. As $\pi \to 0$, we have $\rho \to 0$, $A \to A(0) > 0$ and $\hat{p} \to \hat{p}(\varphi q_L) > 0$, so $D > 0$ near 0. On $\{\hat{p} > 0\}$, $\hat{p}$ is increasing for all three models; under D2 and KR use $p > 1 - 1/\eta'$ there. So $\hat{p}(\varphi q_H) > 0$. Also $A(1) \geq A(0) > 0$, and $\rho \to 0$ as $\pi \to 1$, so $D > 0$ near 1.

At 1/2. $A(1/2)$ does not depend on a_1, while $\rho(1/2) = a_1 \cdot c_0$ with $c_0 > 0$. So $D(1/2) < 0$ once $a_1 > \bar{a} := \max\{0, A(1/2)/c_0\}$.

Large R. If $V_{D,L} > U$, which holds for large R when $\delta_D h_L^D C_L > 0$, both classes favour asking, so the plan is "ask" at every belief. If $\delta_D = 0$ it is "cut" everywhere. Then

$$a_1 = R\,(\bar{c}_H - \bar{c}_L) + \text{const}, \qquad \bar{c}_z = h_z C_z + (1 - h_z)\, q_{D,z}\, \delta_D\, h_z^D C_z,$$

with "ask" and $\bar{c}_z = h_z C_z$ with "cut". In both cases $\bar{c}_H > \bar{c}_L$, because every factor is ordered. □

Example 6. Take $C = (0.5, 0.6)$, $q = (0.5, 0.95)$, $q_D = (0.5, 0.9)$, $\bar{h} = 1$, $U = 100$ and $\delta_D = 0.9$. The last step has $v = 40$, $k = 10$ and one rating.

- D1 with $\eta = 0.5$, $R = 1000$: the agent asks on $[0, 0.0404] \cup [0.6904, 1]$.
- Same, $R = 3000$: the agent asks on $[0, 0.0103] \cup [0.9003, 1]$.
- D2 with $\eta = 0.8$, or KR with $\eta(\Lambda - 1) = 0.8$, $R = 3000$: the agent asks on $[0, 0.0154] \cup [0.9328, 1]$.

The Phase-C plan is "ask" at every belief. Without a sting the agent asks everywhere.

The intuition is that a refusal tells the agent most when they are most unsure. Near $\pi = 0$ a no confirms what they expected; near $\pi = 1$ it is unlikely and barely moves them. In between, a no is likely enough and informative enough that, with a large stake, its expected sting outweighs a step that is worth having for its own sake. This is a new mechanism; it does not arise in the benchmark, where refusals carry no news.

For the guide, a single threshold score per step (Table 5 of the proposal) is not enough. Under any sting the card may need a low asking band and a high one. Both shrink towards the ends as R grows, consistent with Proposition 7 of the proposal. M4 nests M3 ($\psi_n \equiv 1$, $\theta_n \equiv 0$), so both counterexamples apply there too.

3 Conjecture 2: relationship value with commitment

Theorem 7 (First part, at certainty)**.** *Take M4 under S1, with* $h_H(n)$ *nondecreasing in* n*. Stings are arbitrary, on the day and in Phase C, and the budget is arbitrary. At* $\pi = 1$*, the gain from a yes,* G_j*, is nondecreasing in* R *at every step. Whenever* $\hat{p}_j > 0$*, so is the value of escalating* D_j*. Hence the set of* R *at which a known high-chemistry match is asked to escalate is an upper set.*

Proof. At $\pi = 1$ nothing is learned, $p = \varphi_j q_H$, and a no ends the date at the step already reached, so $L = 0$. In Phase C the plan maximises the value against class H,

$$X_H(R) = U + \hat{p}(q_{D,H})^+ \big(V_{D,H}(R) - U\big)^+,$$

which does not depend on n. Hence

$$V_{\text{end}}(1; m; R) = X_H(R) + h_H(m)\big(C_H R - X_H(R)\big).$$

X_H is convex and piecewise linear with slope at most $\delta_D h_H^D C_H \le C_H$, so $C_H R - X_H(R)$ is nondecreasing. So $V_{\text{end}}(1; m; \cdot)$ has increasing differences in (m, R).

As in the first note's Lemma 3, $V_j(1; n) = V_{\text{end}}(1; n) + \hat{p}_j^+ G_j^+$, with

$$G_j = \sup_s \big[W_j^* - c_j\big] + \big[V_{\text{end}}(1; n+1) - V_{\text{end}}(1; n)\big] + \hat{p}_{j+1}^+ G_{j+1}^+.$$

The middle bracket is nondecreasing in R, and G_{j+1} is nondecreasing by induction. The base case is the last step, where the last term is absent. Finally, $D = pG$ for $G < 0$ and $D = \hat{p}G$ for $G \ge 0$, which is nondecreasing in G when $\hat{p} \ge 0$. □

Proposition 8 (First part, short of certainty)**.** *Under S1 and S2, take the last step after n steps, a belief $\pi \in (0, 1]$, and suppose $\delta_D h_L^D C_L > 0$ (or $\delta_D = 0$). Let*

$$\bar{c}_z(m) = h_z(m) C_z + (1 - h_z(m))\, q_{D,z}\, \delta_D\, h_z^D C_z, \qquad \bar{K}(\pi) = \sum_z (\pi_{\text{yes}})_z \big[\bar{c}_z(n+1) - \bar{c}_z(n)\big].$$

Assume $\bar{c}_H(n) > \bar{c}_L(n)$. Then there is $\bar{R}(\pi)$ such that, for $R \ge \bar{R}$, $D(\pi; R)$ is affine in R with slope

$$\Phi(\pi) = \hat{p}(p)\, \bar{K}(\pi) \;-\; \kappa(p)\, (\pi_{\text{yes}} - \pi_{\text{no}}) \big[\bar{c}_H(n) - \bar{c}_L(n)\big]. \tag{4}$$

(i) $\Phi(1) = \hat{p}(\varphi q_H)\big[\bar{c}_H(n+1) - \bar{c}_H(n)\big] \ge 0$. If this is positive, $\Phi > 0$ on a neighbourhood of $\pi = 1$, and Conjecture 2(a) holds there for all large R.

(ii) Where $\Phi(\pi) < 0$, a higher R lowers the value of escalating, and the agent stops asking for large R. Proposition 7 of the proposal then survives commitment.

Proof. For large R the plan is "ask" at every belief, as in Proposition 5. Then

$$V_{\text{end}}(\pi; m) = R \sum_z \pi_z \bar{c}_z(m) + \sum_z \pi_z e_z(m),$$

with e_z free of R. The rating is a martingale on this affine function, so

$$G = W^* - c + V_{\text{end}}(\pi_{\text{yes}}; n+1) - V_{\text{end}}(\pi; n), \qquad L = V_{\text{end}}(\pi; n) - V_{\text{end}}(\pi_{\text{no}}; n).$$

$\Delta = G + L$ has R-coefficient $\sum \pi_{\text{yes}} \bar{c}(n+1) - \sum \pi_{\text{no}} \bar{c}(n)$, which is positive, so $\Delta > 0$ for large R. $D = pG - (1-p)L - \kappa\Delta$ is then affine in R. Using $p\pi_{\text{yes}} + (1-p)\pi_{\text{no}} = \pi$, the slope of $pG - (1-p)L$ is $p\bar{K}$, and the slope of $\kappa\Delta$ is

$$\kappa\big[\bar{K} + (\pi_{\text{yes}} - \pi_{\text{no}})(\bar{c}_H(n) - \bar{c}_L(n))\big].$$

Subtracting gives (4). Continuity of Φ gives (i). □

Example 9. Take $C = (0.3, 0.8)$, $q = q_D = (0.5, 0.9)$, $\bar{h} = 0.5$, $\delta_D = 0.9$, D1 with $\eta = 0.5$, and a last step with $v = 40$ and $k = 15$.

- Weak bonding, $\psi = 1.05$: at $\pi = 0.9$, $D = 16.81,\ 13.43,\ 0.74,\ -16.17$ for $R = 10^3,\ 5 \cdot 10^3,\ 2 \cdot 10^4,\ 4 \cdot 10^4$, and $\Phi(0.9) = -0.0008$. At $\pi = 1$, D rises: 28.3, 63.0, 193.1, 366.5.
- Strong bonding, $\psi = 1.5$: $\Phi(0.9) = 0.068 > 0$.

The numerical slopes equal (4) to six decimals.

So "likely high-chemistry" must be quantified. A higher R helps escalation exactly where $\Phi > 0$, that is, where commitment outweighs the sting of the news a refusal carries. That region always contains a neighbourhood of certainty, and can be small.

Proposition 10 (Second part)**.** *Take sliding, S1 and S2, let $K(\pi; R) = V_{\text{end}}(\pi; n+1) - V_{\text{end}}(\pi; n)$, and let $\Delta\theta = \theta_{n+1} - \theta_n > 0$.*

(i) Suppose at π the optimal plan is the same plan d at n and at $n+1$, for every R in $[R_1, R_2]$. Then

$$K(\pi; R) = \Delta\theta \sum_z \pi_z \big(C_z R - X_z^d(R)\big)$$

is nondecreasing on $[R_1, R_2]$. So if π has non-negative commitment value at R_1, it keeps it.

(ii) At $\pi = 0$ and $\pi = 1$ this holds for all R. If there is no second-date option (plan "cut"), the negative set is

$$N(R) = \Big[0,\ \frac{U - C_L R}{(C_H - C_L)R}\Big),$$

which shrinks as R rises.

(iii) In general the conjecture is false. Take $C = (0.28, 0.93)$, $q_D = (0.71, 0.99)$, $\bar{h} = 0.85$, $\delta_D = 0.95$, $U = 100$ and $\theta_{n+1} - \theta_n = 0.06$. At $\pi = 0.27$, $K = +0.271$ at $R = 230$ but $K = -0.362$ at $R = 250$.

Proof. (i) By Lemma 1, with the plan fixed, K is as displayed. Moreover $\partial X_z^{\text{ask}}/\partial R = q_{D,z}\delta_D h_z^D C_z \le C_z$, and $X_z^{\text{cut}} = U$.

(ii) At a degenerate belief, π_{not} is the same degenerate belief, so the plan maximises X_z^d and does not depend on n. Then

$$C_z R - \max_d X_z^d = \min_d \big(C_z R - X_z^d\big)$$

is nondecreasing.

(iii) Computed in `verify_v2.py`. At $R = 230$ the agent would ask for another date after a date that reached n steps, but cut after one that reached $n+1$: sliding makes "no relationship" worse news about chemistry. At $R = 250$ the agent asks after both. Commitment then locks in a relationship that, at this belief, is worth less than the second-date plan ($\sum \pi_z (C_z R - X_z^{\text{ask}}) < 0$). □

Remark 11. In a targeted search the largest violation was 0.27 on a scale where $U = 100$. Conjecture 2(b) is therefore close to true, but fails exactly where the after-date plan switches between n and $n+1$.

4 Conjecture 3 in the several-dates model

Conjecture 3 is stated for M2+, the full date-by-date recursion. Here M2+ is built on the chemistry prior, which keeps the belief one-dimensional and the after-date update closed form. Talking between dates is switched off; take c_{talk} large. A date starts with its opening step, then asks; after it comes a relationship, or Phase C, where another date is worth $\delta_D V_{\text{date}}(\pi_{\text{yes}})$. The fixed point V_{date} is computed by value iteration on a grid of π; the contraction factor is at most δ_D.

Proposition 12. *(a)* Phase C (true). *With no sting anywhere, the set of beliefs at which the agent asks for another date shrinks as U rises.*

(b) On the day (false). *Take $C = (0.24, 0.63)$, $q = (0.74, 0.76)$, $q_D = (0.42, 0.59)$, $\bar{h} = 0.36$, $R = 1470$ and $\delta_D = 0.88$. The opening step is coffee ($v = 10$, $k = 8$, one rating); the second step has $v = 96$, $k = 50$ and two ratings. There is no sting.*

- *The on-day asking set is a threshold at both $U = 313$ and $U = 322$.*
- *Raising U* lowers *the threshold, from 0.6272 to 0.6119. At both thresholds the agent would ask for another date.*
- *The Phase-C asking region moves from $\pi \ge 0.385$ to $\pi \ge 0.406$, as part (a) says.*

The numbers are identical on grids of 401 *and* 1601 *points.*

Proof of (a). With no sting, every policy's value has the form $\alpha + \beta U$. Here $\beta \in [0, 1]$ is the expected discount factor at the time the agent returns to search, and zero if they never do. So $V_{\text{date}}(\pi; U)$ is a supremum of affine functions with slopes in $[0, 1]$. Hence $\delta_D V_{\text{date}}(\pi'; U) - U$ is strictly decreasing in U. In Phase C a no leads to U, so the agent asks iff $\delta_D V_{\text{date}}(\pi_{\text{yes}}) \geq U$. □

The mechanism is the pivotality of the first note. The value of learning on the day comes from the after-date decision, and a better outside option moves that decision towards indifference, which makes learning more valuable. The full recursion does not remove this. The version of M2+ with independent Beta priors is not covered, because "no relationship" breaks conjugacy there; the mechanism is the same.

5 Conjectures 4 and 5, and the value of information

The statements of Conjectures 4 and 5, and of Prediction 7 of Appendix A.9, are unchanged, and the first note's results apply. Two points are specific to the chemistry prior.

Conjecture 4. The first note shows that thresholds rise as the budget falls in the core model. That carries over to M3 and M4 whenever refusals are weakly bad news ($L \geq 0$). If $\hat{p} \geq 0$, the value of asking is nondecreasing in the value after a yes, which grows with the budget. If $\hat{p} < 0$ and $L \geq 0$, then $D = \hat{p}\,\Delta - L < 0$ (or $D = pG - L \leq 0$) at every budget, so the agent never asks.

Conjecture 5 and negative information value. The first note's guarantee that spend is never below Proposition 1's level rests on information about compatibility having non-negative value. Under the chemistry prior with a sting this fails. Every signal about the class also moves the chance of a yes. At the last step, with affine V_{end} and D1,

$$D = pA - \eta(1-p)A - \eta a_1(\pi_{\text{yes}} - \pi),$$

using $(1-p)(\pi_{\text{yes}} - \pi_{\text{no}}) = \pi_{\text{yes}} - \pi$. The first term is a perspective of a convex function, and the last is convex. The middle term, $-\eta(1-p)\,W^*(\pi_{\text{yes}})$, can be concave.

Example 13. Take $C = (0.11, 0.6)$, $q = (0.81, 0.88)$, $q_D = (0.5, 0.9)$, $\bar{h} = 0.5$, $R = 129$, $U = 100$ and $\delta_D = 0$. The last step has $v = 31$ and $k = 10$, with D1 and $\eta = 1.5$. A binary cue with $P(\text{enc} \mid H) = 0.52$ and $P(\text{enc} \mid L) = 0.48$ is MLRP. Observed at $\pi = 0.44$, it has EVSI $= -0.0249$. The agent asks at the prior and at both posteriors. Without the sting, the same cue's EVSI is $+0.0037$.

So in M3 with a sting, neither the non-negativity of information value nor the bound "spend is never below Proposition 1's level" is guaranteed. Without a sting both hold, because the value is then convex in π. This answers, for the chemistry prior, the question Section 4.4 of the proposal still lists as open. Section 10.1 already withdraws the claim that the value of information can be negative; it should be reinstated in this qualified form.

6 Other points on the revised proposal

1. *Proposition 6 needs $C_z R \geq X_z^d$, or a slope condition.* Take $C = (0.2, 0.8)$, $h = (0.1, 0.5)$, $R = 100$, $U = 200$, and the plan "cut", which pays U against both classes. Then $V_{\text{end}}(0) = 182 > V_{\text{end}}(1) = 140$. V_{end} falls in π and a refusal is *good* news: a likely relationship is worth less than search.

2. *Table 3 and Prediction 6.* The benchmark's investment motive is not strongest "where beliefs are least settled". The first note shows it peaks near the after-date indifference point and can be zero at maximal uncertainty, which is also what the proposal's own Proposition 4 implies.

3. *Table 5.* "One threshold score per step" should become "up to two asking bands per step" wherever a sting is used (Proposition 5). The sting check of Section 6.1 will then flag exactly the middle band.

4. *Conjecture 2.* Restate "for likely high-chemistry matches" as "where $\Phi > 0$" (Proposition 8). Restate the second part as holding where the after-date plan does not switch with the step reached.

5. *Conjecture 3.* Keep the Phase C part, and replace the on-day part by the first note's pivotality lemma: thresholds rise with U where the agent would cut contact, and fall where they would ask for another date.

Numerical verification

`python3 verify_v2.py` (about one minute) reproduces every number above. It also runs randomised checks of:

- Theorem 3, as single crossing at every step in 150 models;
- Theorem 7, in 150 M4 models with random stings, bonding or sliding;
- the rule (3) and formula (1);
- formula (4);
- Proposition 10(i)–(ii), over 115,000 parameter pairs;
- Proposition 12(a), in random M2+ models.

The output is in `verify_v2_output.txt`.